

Perovskite Photovoltaics in January-June 2026: Defect passivation, interface and contact engineering

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Abstract

Evaluation of 3328 publications across 6 months, with 884 examined in depth, reveals that reported efficiency records mask severe mechanistic contradictions and pervasive validation deficits across the discipline. Independent verification remained scarce: only 114 investigations reported an accredited power conversion efficiency (PCE), with the certified frontier reaching 34.85% in monolithic tandems and 27.6% in single-junction cells. Rather than advancing across the observation window, certified performance remained confined within a narrow efficiency corridor where contradictory physical models yielded indistinguishable macroscopic photovoltaic parameters. These certified benchmarks were attained through competing interfacial and bulk defect mitigation strategies. At top interfaces, reports diverged sharply on whether non-radiative recombination suppression requires reconstructing low-dimensional capping phases or whether such layers impose deleterious carrier extraction bottlenecks. At buried contacts, self-assembled monolayer (SAM) engineering alternated between static electrostatic dipole alignment using polar co-adsorbents and mitigating dynamic interfacial chemical redox reactions, while tandem subcell coupling contested whether current matching demands laterally conductive transparent conducting oxides or direct tin oxide recombination layers across textured interfaces. Concurrently, formamidinium lead triiodide phase stabilization conflicted between non-destructive contact-triggered templating without interdiffusion and active cation exchange driven by Ostwald ripening. Device scaling imposed an uncompromising area penalty, with the maximum certified aperture reaching 655.2 cm^2 at an efficiency of 26.8%. This scaling loss stems from conflicting precursor-solvent coordination kinetics during blade and slot-die coating, where suppressing intermediate solvent adducts conflicts with Lewis-base coordination required to retard premature nucleation, compounded by drying non-uniformities, localized sheet resistance penalties, and pinhole multiplication. Operational durability evidence is severely deficient across the field: only 39 publications reported stability lifetimes to the defined retention threshold, with the longest reaching 4100 h. Evidence for long-term operational resilience remains thin, as operando structural characterization demonstrates that bulky passivators designed to relieve lattice strain induce localized lattice expansion and mechanochemical fatigue under thermal stress, while buried SAM contacts degrade through unresolved competition between field-driven lateral aggregation into isolated clusters and out-

of-plane molecular desorption. Reporting practices reinforce this empirical deficit: only 39 studies implemented standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. Abstract-level text extraction confirms that while 52.1% of publications stated an efficiency, only 8.5% reported independent certification, showing that most claims rely on uncertified dynamic sweeps without aperture masking verification, spectral mismatch calibration, or steady-state maximum power point tracking (MPPT). Mechanistic synthesis indicates that apparent efficiency advances frequently reflect transient electrostatic screening and interfacial dipole realignment rather than permanent electronic defect remediation. Because diametrically opposed physical models produce indistinguishable macroscopic parameters, standard current-voltage characterization cannot separate interfacial recombination velocity from bulk trap states. Resolving whether operational degradation proceeds through interfacial redox, lateral contact dewetting, or mobile point defect accumulation requires operando, spatially resolved transient spectroscopy paired with depth-resolved photoemission under simultaneous electrical bias, continuous illumination, and elevated temperature, alongside standardized ISOS aging on certified apertures.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite; tandem solar cells; operational stability

1. Introduction

Recent investigations into interfacial decay in metal halide perovskite solar cells arrive at fundamentally contradictory conclusions regarding the primary driver of non-radiative recombination and operational degradation. While one body of literature attributes performance loss at the buried contact to chemical redox reactions between migrating iodide species and adjacent charge extraction layers, competing reports argue that performance decay is governed entirely by field-driven accumulation of mobile point defects that screen the built-in potential. Disentangling these mechanisms requires operando, spatially resolved transient optoelectronic spectroscopy combined with depth-profiled X-ray photoemission under simultaneous electrical bias and continuous illumination. This unresolved mechanistic conflict reflects a broader transition across the discipline. The principal frontier of perovskite photovoltaics has progressed beyond chasing isolated power conversion efficiency (PCE) records on sub-0.1 cm² laboratory apertures. Instead, the central challenges reside in reproducible third-party verification, maintaining performance across commercially relevant module areas, and ensuring operating lifetime under continuous operational stress.

Across the six months from January to June 2026, 3328 indexed works met the scope gate and 884 were read closely. This body of literature reveals that closing the gap between small-area laboratory devices and commercial modules requires resolving several interrelated physical processes across the device stack. This review follows seven critical mechanistic threads: trap states, passivation and ionic motion; contact chemistry at the buried interface; A-site and halide composition engineering; tandem architecture and current matching; degradation mechanisms under operation; and large-area coating and the area penalty. Defect chemistry cannot be isolated from ionic dynamics, as shallow point defects migrate under operating electric fields, transforming into deep carrier traps or accumulating at transport layers. Similarly, contact chemistry at the buried interface dictates initial crystallization kinetics, internal strain, and energetic alignment, which dictate long-term resistance to ion migration and delamination. Meanwhile, A-site and halide composition engineering directly governs thermodynamic phase stability against segregation under light and thermal stress, setting the operational boundary for tandem architecture and current matching in monolithic multijunctions.

Translating these microscopic stabilization strategies to industrially scalable dimensions exposes the area penalty incurred during large-area coating, where meniscus drying kinetics, fluid dynamics, and substrate thermal gradients induce spatial variations in film stoichiometry and defect density. Under actual operating conditions, these initial microstructural non-uniformities accelerate localized degradation mechanisms under operation, precipitating failure through moisture ingress, heat-induced halide redistribution, and electrode corrosion. Evaluating these interconnected phenomena across a six-month window provides a necessary diagnostic perspective: a single month captures only isolated values and transient records, whereas six months reveal whether reported performance metrics actually moved or stagnated. A half-year view cannot resolve multi-thousand-hour degradation trajectories or guarantee twenty-year service lifetimes, but it offers sufficient temporal depth to distinguish genuine thermodynamic stabilization from statistical noise and short-lived laboratory anomalies.

2. The Certified Frontier Across January-June 2026

A fundamental disagreement governs recent two-terminal (2T) tandem reports concerning the origin of non-radiative voltage deficits: while one investigation attributes performance limits to defective hole-extraction dynamics at the buried self-assembled monolayer (SAM) contact [1], another concludes that phase segregation and deep trap formation within the bulk wide-bandgap perovskite dominate carrier recombination [2]. Resolving this conflict requires *operando* absolute photoluminescence quantum yield (PLQY) imaging coupled with transient absorption spectroscopy under external bias, which can isolate interfacial recombination velocity from bulk quasi-Fermi level splitting collapse. These opposing mechanistic attributions yielded certified power conversion efficiency (PCE) values of 33.6% for contact optimization [1] and 32.95% for bulk defect mitigation [2]. Across the six-month period, independently certified 2T tandem efficiencies occupied a narrow corridor spanning 31.93% [3] to 34.85% [4], illustrating that divergent physical hypotheses are routinely invoked to explain marginal variations within the same narrow efficiency envelope.

Over the half year, the best certified PCE moved from 29.2% in January to 33.6% in June, representing a change of 4.4 percentage points. Rather than demonstrating an advancing technological frontier, this six-point monthly sequence reflects the stochastic noise of manuscript indexing schedules. The recorded progression of monthly certified peaks from 29.2% in January, to 32.46% in February [5], 31.93% in March [3], 32.81% in April [6], 33.5% in May [7], and 33.6% in June [1] failed to establish a newly discovered physical regime. The highest independently certified efficiency of the entire half year, 34.85% [4], appeared in March and exceeded every certified value reported in May and June. Consequently, the certified ceiling did not move outward, but repeated within established statistical bounds. Every leading certified result across this period was achieved on multi-junction devices, specifically 2T tandems, rather than single-junction absorbers. The verified boundary therefore reflects subcell current matching, optical coupling, and recombination-junction losses rather than fundamental breakthroughs in single-junction materials science.

A persistent gap separates uncertified claims from verified physical metrics: only 114 of 872 closely read papers reported an independently certified efficiency. In January, 15 papers reported certified values out of 475 indexed works, with the top self-reported efficiency of 31.85% standing near the certified 29.2%. In February, 17 papers reported certified values out of 499 indexed works, where self-reported metrics reached 32.73% alongside an independently certified 32.46% [5]. March included 14 certified papers among 559 indexed works, recording a best self-reported 32.43% against a certified 31.93% [3]. In the second quarter, uncertified self-reported claims detached completely from verified

performance, escalating to 41.6% in April across 589 indexed works (21 reporting certified values) and 44.36% in May across 583 indexed works (29 reporting certified values). By contrast, independently certified 2T tandem efficiencies converged into a narrow band, supported by shared interface passivation and additive treatments: April yielded certified values of 32.81% [6] and 32.45% [8], while May produced a dense cluster with 33.5% [7], 33.48% [9], 33.37% [10], 33.04% [11], and 32.1% [12]. June recorded 14 certified reports among 623 indexed works, where self-reported metrics of 33.59% fell slightly below certified results at 33.6% [1] and 32.95% [2]. Independent certifications clustered tightly across these studies [7,9,10,11,12], confirming that uncertified claims above 40% arise from spectral mismatch errors or omitted aperture masking rather than verified material mechanisms.

The concentration of high efficiencies in small-area cells exposes the physical barrier of module scaling, where geometric parasitic losses depress performance. Across the half year, the share of closely read papers expanded from January (12.6%), February (15.0%), and March (16.7%) through April (18.8%), May (17.7%), and June (19.2%), reflecting increased publication activity around scalable manufacturing. However, active area expansion introduces severe lateral sheet resistance in transparent conductive oxide (transparent conducting oxide (TCO)) electrodes, subcell interconnect dead zones, and non-uniform film drying, which systematically degrade the fill factor (FF). At 100 cm², certified module efficiencies were limited to 26.59% [13] and 24.7% [14], where localized series resistance and laser-scribed interconnect defects constrain power extraction [13,14]. At larger areas, efficiency diminished further to 27.12% across 655.2 cm² [15], 26.62% across 651 cm² [16], and 23.3% across 694.5 cm² [17]. At the extreme of reporting, a 900 cm² device was presented without a certified efficiency [18]. These multi-junction and module data demonstrate that verified small-cell records above 33% fail to translate directly into scalable technology, where macroscopic transport resistance and deposition non-uniformities supersede nanometer-scale interface engineering.

3. Trap States, Passivation and Ionic Motion

Recent investigations into trap passivation and ionic transport diverge sharply regarding the structural configuration required to deactivate surface defects without impeding charge collection. One faction asserts that capping the three-dimensional (3D) perovskite absorber with low-dimensional phases is essential to eliminate residual reagents and suppress non-radiative recombination. Vapor-mediated dimensional reconstruction using triethylamine-pentafluorophenol (TEA-PFP) to react with excess lead iodide (PbI₂) forms a one-dimensional (1D) TEAPbI₃ interfacial phase, yielding a certified power conversion efficiency (PCE) of 26.15% (champion 26.71% on 1 cm²) and a fill factor (FF) of 89.13% [19]. In situ generation of a guanidinium derivative similarly templates a two-dimensional/three-dimensional (2D/3D) heterostructure, giving a certified PCE of 26.2% (champion 26.49%) with 95% retention after 1000 h of maximum power point tracking (MPPT) [20]. Conversely, competing reports argue that low-dimensional capping phases introduce anisotropic transport barriers, carrier accumulation, and phase instability under operational bias. They demonstrate instead that non-2D organic co-crystal layers passivate defects via reinforced hydrogen-bonding networks and enhanced electron mobility, achieving a champion PCE of 26.35% with a T80 lifetime of 1392 h and 88% efficiency retention after 2040 h at approximately 50% relative humidity [21]. This rejection of low-dimensional capping is corroborated by stereoelectronic ligand engineering: pyridine or pyrimidine rings forming dual lead-nitrogen (Pb-N) coordination and Pb-I- π interactions drive planar ligand alignment, producing atomic-scale defect mitigation with sub-nanometre charge transfer and a certified PCE of 27.41% (stabilised at 26.85%) without dimensional reconstruction [22]. Settling this dispute requires operando bias-dependent, depth-resolved grazing-incidence wide-angle X-ray scattering paired

with local transient absorption microscopy under continuous thermal stress to establish whether 1D and 2D phases degrade into resistive barriers under operational electric fields or preserve a stable passivation contact.

Chemical passivation strategies have targeted undercoordinated metal cations, halide vacancies, and antisite defects through coordinated Lewis base and acid interactions. Carboxyl-group chelation of undercoordinated lead (Pb^{2+}) by dipicolinic acid (DPA) passivates surface and bulk traps in 1.68 eV wide-bandgap (WBG) cells, reaching 22.91% PCE and a champion 31.66% in two-terminal (2T) perovskite/silicon tandems [23]. Lewis acid-base coordination using 3-hydroxyflavone coordinates Pb^{2+} and increases the ion-migration barrier, though its champion PCE of 26.6% was measured on an uncertified, small area of 0.09 cm^2 [24]. Adding meta-fluoro-phenylethylammonium iodide (m-F-PEAI) to ethyl acetate antisolvent employs the fluorine atom as a Lewis base toward Pb^{2+} while establishing directed hydrogen bonds with formamidinium iodide (FAI) to retard crystallization, achieving 26.41% champion PCE [25]. Redox-active dabrafenib (DRFN) coordinates multiple Pb^{2+} centers via its sulfonyl group and heterocyclic nitrogen, achieving peak PCEs of 25.28% in n-i-p and 26.62% in p-i-n devices [26]. Molecular dipole passivators 2-amino-4,6-dichlorotriazine and 1-adamantanecarbonyl chloride hydrochloride (ACHCl) combine Lewis-base coordination with chloride filling of iodine vacancies (VI), correcting PbI_6 octahedral distortion and reducing recombination to reach champion PCEs of 26.58% and 26.12%, respectively [27,28]. Multi-defect healing additives pairing pentafluorophenol (PFP) with 1,4-dicyanobenzene (DCB) target VI, lead vacancies (VPb), and lead-on-iodine (PbI) antisites, retaining 95% efficiency after 1300 h at 65 °C and 94% after 1000 h of 1-sun MPPT (champion PCE 26.47%) [29]. Complementary bimolecular passivation with 1,4-phenylenediammonium diiodide (PDADI) and 11-mercaptoundecanoic acid (MUA) reduces trap density from 3.00×10^{15} to $1.65 \times 10^{15}\text{ cm}^{-3}$ (champion PCE 26.15%) [30], while quantitative inkjet printing of PDADI alone yields 26.1% PCE [31]. In the bulk, precursor dicyandiamide improves vacancy formation energies and crystallinity, producing a monolithic perovskite/silicon tandem champion PCE of 32.03% with an open-circuit voltage (V_{oc}) of 1.95 V [32]. Chiral configurations show that stereochemical orientation controls defect thermodynamics: (R)-(+)- α -phenylethylamine (R-Pye) exhibits higher binding energy (9.45 versus 7.33 eV) and increased VI formation energy (4.56 versus 2.46 eV) than (S)-(-)- α -phenylethylamine (S-Pye), reaching 26.13% champion PCE [33]. To prevent ammonium deprotonation, sterically protected cations like N,N-dimethylcyclohexylammonium iodide (DMCHAI) inhibit side reactions with formamidinium (FA+), maintaining over 86% efficiency after 550 h at 85 °C (champion PCE 26.06%) [34], matching steric and electronic tuning in trimetazidine (TMZ) that achieves 26.07% PCE with over 90% retention after 1400 h [35]. Latent passivator 3-trifluoromethylphenylboronic acid pinacol ester (3-CF₃-PhBpin) hydrolyzes in situ into 3-CF₃-PhB(OH)₂ to passivate crystallization defects, achieving 26.25% in rigid and 24.80% in flexible n-i-p cells [36]. Machine-learning frameworks integrating domain-knowledge large language models with Bayesian optimization have independently proposed untested four-component recipes (3MTPAI, PDAI2, EDAI2, and PipDI) reaching 26.0% champion PCE [37].

Over the six-month period, investigation of ion migration and halide segregation revealed an incompatibility between static immobilization and dynamic strain dissipation. Static strategies pin mobile ions through spatial confinement or covalent polymerization. Molecular confinement domains suppress halide segregation by constraining FA+ motion, enabling single-junction efficiencies of 24.21% (1.68 eV) and 21.20% (1.78 eV) and a 33.59% champion 2T tandem PCE [38]. Incorporating phenylethylguanidinium (PEGA+) salts suppresses cation drift in 1.55 eV perovskites, exceeding 26.2% PCE with over 96% retention after 1100 h and 89% after 800 h at 85 °C [39]. In situ poly-

merization of vinylphosphonic acid outclasses small-molecule organoammoniums, retaining over 90% PCE after 1600 h of 1-sun MPPT (certified 26.24%, champion 26.54%) [40]. In situ polymerization of sodium thiocetate (TANA) combined with sodium interstitial doping regulates crystallization and suppresses ion migration, retaining 92.3% efficiency after 1000 h under International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-2I conditions (certified 26.75%, champion 26.9% on a 10.24 cm² module) [41]. Self-polymerizable fluorinated additives (PFPA) achieve a champion PCE of 26.01% on a 14 cm² area, retaining 90.0% efficiency after 840 h at 85 °C in nitrogen and 95.1% after 1000 h of MPPT [42]. Room-temperature condensation between amino and acyl chloride groups establishes a dual-anchor polymeric interlayer, preserving 85% efficiency after 2300 h of MPPT and 1680 h at 65 °C (champion PCE 26.12%) [43]. At the transport interface, bisphosphonate interlayers (2OBCz) between the perovskite and hole transport layer (hole-transport layer (HTL)) inhibit ion migration, retaining 90.7% efficiency after 1000 h under ISOS-D-2 conditions compared to monophosphonates (LBCz) (champion PCE 26.02%) [44]. In narrow-bandgap tin-lead (Sn-Pb) cells, formamidinium bisulfate (FBH) dipole management reverses top-surface grain-boundary states to induce downward band bending, preserving over 80% efficiency after 400 h at 85 °C and 90% after 1872 h in a glovebox (champion PCE 23.25%, tandem 29.67%) [45].

In direct conflict with the static immobilization premise, evidence indicates that rigid polymer networks suffer fatigue and microcracking under thermal cycling, opening pathways for ion accumulation. Dynamic networks resolve this failure mode: mechanochemically reinforced dual-dynamic covalent seeding (DDS) via orthogonal Diels-Alder and oxime-carbamate reactions constructs an interpenetrating covalent mesh at grain boundaries that heals under strain while passivating defects and blocking ion and moisture ingress (certified PCE 26.69%, champion 26.95%, ISOS-L-2I) [46]. Integrating photoswitchable isomers at grain boundaries suppresses lattice bond rupture and defect accumulation during cyclic illumination and thermal fluctuation, retaining over 95% of initial efficiency after 2,0 h of UV-containing light cycling at 65 °C and 500 thermal cycles between -40 °C and 85 °C (certified PCE 26.9%, champion 27.2%) [47]. Dynamic methyl viologen (MV) interlayers similarly maintain 90% efficiency after 1,0 h of AM 1.5G MPPT (champion PCE 26.44%, Voc 1.167 V, FF 85%) [48]. Systematic temperature-dependent transient ion drift measurements across sustained thermal cycles are necessary to determine whether dynamic covalent networks surpass rigid polymers in preventing long-term mobile ion accumulation.

Surface passivation and ionic stabilization depend heavily on energetic alignment across selective contacts. In inverted p-i-n devices, modulating intermolecular N-H...N hydrogen bonding between two passivators tunes energy-level offsets, yielding a 24.04% wide-bandgap single-junction PCE and a certified monolithic 2T tandem PCE of 33.48% (champion 33.63% on 1 cm²) [9]. Self-assembled monolayers (SAMs) on nickel oxide (NiO_x) suffer from phase aggregation and incomplete coverage. Co-adsorbing N-indoleacetic acid with [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Ph-4PACz) forms phase-homogeneous binary monolayers via pi-pi stacking and hydrogen bonding, achieving 26.95% PCE (certified 26.57%) on 0.0717 cm² and 25.61% on 1 cm² [49]. Blending 4PABCz with Me-4PACz forms a Co-SAM that enhances coverage on NiO_x, passivating bottom defects and suppressing non-radiative recombination (champion PCE 26.33%) [50]. Positional isomerism provides structural control: selective chlorine placement on the peripheral phenyl ring of Ph-4PACz (ortho-, meta-, and para-chlorinated isomers) enables lattice-matched dual-site passivation (certified PCE 26.13%, champion 26.58%) [51]. Indolocarbazole-based SAMs (M3PAICz-1) demonstrate versatility across bandgaps, delivering 26.12% PCE in 1.55 eV cells and 22.19% in 1.68 eV wide-bandgap devices [52], while tripodal phosphonic acids anchor multidentately with face-on pi-stacking and segregate to

grain boundaries to passivate undercoordinated Pb^{2+} (certified PCE 26.35%) [53]. The underlying metal-oxide contact can also be chemically stabilized: bromoacetamide (BAA) undergoes in situ dissociative adsorption on sputtered NiO_x , where bromide ions coordinate surface nickel to stabilize Ni^{3+} and suppress disorder (certified PCE 26.07%, champion 26.31%) [54].

At the electron-selective contact, isomer-free multi-adduct fullerene derivatives $\text{C}_{60}(\text{NHR})_{40}$ (TATPC) blended into [6]-phenyl-C61-butyric acid methyl ester (PCBM) raise the lowest unoccupied molecular orbital (LUMO) level, strengthening interfacial interactions to reach 26.66% champion PCE on a 1.04 cm^2 aperture [55]. Fused-ring electron acceptor (FREA) additives similarly passivate defects and accelerate electron extraction, yielding a certified PCE of 26.35% (champion 26.66%) with under 8% degradation after 2150 h under continuous illumination [56]. In tandem subcells, differential coordination using 1,10-phenanthroline coordinates Sn^{2+} more strongly than Pb^{2+} , synchronizing crystallization to deliver a Sn-Pb single-junction V_{oc} of 0.887 V, an FF of 81.27%, and a PCE of 23.63% (tandem champion 29.08%) [57]. Dual-molecule precursor additives (AGS and AHC) similarly coordinate Sn-Pb crystallization, achieving 23.44% single-junction efficiency and a certified tandem PCE of 28.57% (champion 29.36%) [58], while halide-cation interactions tuned via chlorine and fluorine salts yield wide-bandgap subcells of 20.5% and 20.3% PCE, retaining 87% tandem efficiency after 700 h of MPPT (tandem champion 29.3%) [59]. Interface passivation extends to the silicon bottom cell; dark annealing of n-type tunnel oxide passivated contact (n-TOPCon) silicon cells at 300 °C reduces defect-assisted recombination, raising subcell efficiency from 18.92% to 20.23% in four-terminal tandems with a 26.75% champion PCE [60]. In comparison, idealized one-dimensional numerical simulations (SCAPS-1D) for AgCdF_3 absorbers with transition-metal dichalcogenide transport layers demonstrate that reducing bulk defect density to $1 \times 10^{13}\text{ cm}^{-3}$ at $0.4\text{ }\mu\text{m}$ thickness defines an efficiency benchmark of 31.92% [61].

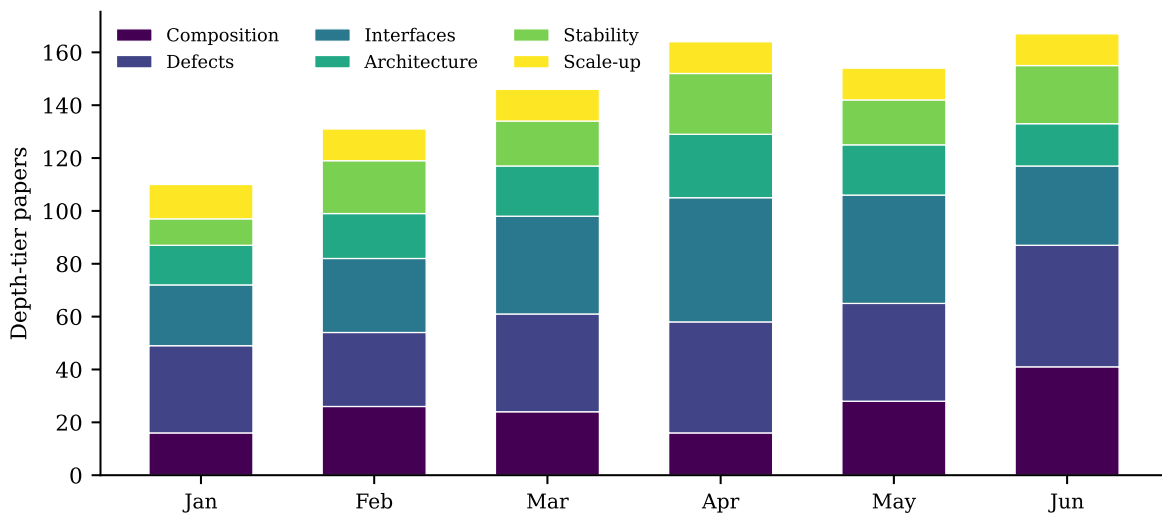


Figure 1. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Contact Chemistry at the Buried Interface

A fundamental contradiction defines recent investigations into carrier transfer across self-assembled monolayer (SAM) contacts at the buried interface of perovskite solar cells (PSCs). On one side,

prevailing models treat hole extraction as a static electrostatic problem governed by ground-state vacuum work functions and interfacial dipole alignment [62,63,64]. In this framework, introducing polar co-adsorbents such as cyanoacetic acid (CA) to raise the work function of [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) yields an uncertified champion power conversion efficiency (PCE) of 26.52% [62], while assembling asymmetric Br-4PhCz monolayers with large dipole moments produces an open-circuit voltage (V_{oc}) of 1.36 V in 1.77 eV wide-bandgap cells, delivering a certified tandem PCE of 28.98% (champion 29.35%) on a small 0.05 cm² area [63]. Similarly, terminal group tuning alone is reported to suppress non-radiative recombination to reach 95% and 91% of the Shockley-Queisser limit in 1.53 eV (uncertified champion PCE of 26.88%) and 1.72 eV devices [64]. Conversely, direct operational interrogation reveals that charge transfer across SAM-modified interfaces is dynamic and governed by a self-limiting negative feedback loop [65]. In this opposing mechanism, photo-injected holes accumulating at the buried contact establish a transient opposing electric field that induces rapid field saturation over short characteristic times, triggering a sharp decay in extraction efficacy regardless of favorable initial band offsets [65]. These two positions cannot be reconciled: if transient space-charge accumulation dominates working conditions, static energy level optimization cannot prevent extraction decline under continuous illumination. Resolving this conflict requires operando, time-resolved Stark-shift spectroscopy or bias-dependent transient electro-absorption measurements under continuous illumination to determine whether static interfacial dipoles persist or undergo severe electrostatic screening during steady-state carrier flux.

Beyond dipolar tuning, modifying electronic coupling across buried hole transport layers (HTLs) has targeted both chemical linker architecture and non-radiative decay channels. Inserting thiophene rings into sulfur-modulated, fully conjugated monolayers (SMe-TPA-ThPA and SMe-TPA-PA) raises inverted device efficiencies from 24.31% in oxygen-linked controls to 26.52% and 25.40% by strengthening intermolecular pi-electron interactions and energetic alignment [66]. Interfacial polarization can also be tuned through meta-substituted bifunctional molecules such as m-APCA, which suppresses SAM aggregation to achieve a certified 26.71% PCE (champion 26.77%) [67], or by introducing a 4-aminobutylphosphonic acid (4-ABPA) molecular bridge that limits interfacial voltage loss to 31 mV, yielding uncertified efficiencies of 25.56% in n-i-p and 26.45% in p-i-n configurations [68]. Rather than relying solely on electrostatic barrier adjustment, resonant energy transfer channels have been established: incorporating the UV-312 molecule introduces a Förster resonance energy transfer (FRET) pathway that facilitates charge separation while suppressing ultraviolet-induced lead-iodide bond dissociation, achieving an uncertified 27.05% PCE with a V_{oc} of 1.186 V on a 0.09 cm² aperture [69]. However, the efficacy of these small-molecule modifiers depends entirely on substrate anchoring density and the absence of microscopic pinholes.

The second physical problem centers on anchoring density, molecular orientation, and phase segregation on rough or chemically heterogeneous substrates. Small-molecule phosphonic acids bind erratically to transparent conducting oxides because of non-uniform hydroxyl distributions. Over the past six months, synthetic strategies shifted from modifying underlying oxides toward catalytic assembly and macromolecular networks. Initial interventions altered substrate surface chemistry directly: depositing three atomic layers of aluminum oxide via atomic layer deposition (ALD) generates dense hydroxyl sites that align self-assembled monolayers (SAMs) vertically, yielding an uncertified 26.63% PCE on 0.09 cm² and 22.97% on a 14.40 cm² mini-module [70]. In two-terminal perovskite/silicon tandems, incorporating carbon into polysilicon tunnel recombination layers elevates surface hydroxyl density to anchor SAMs securely, producing a certified 33.5% tandem PCE (champion 33.84%) on 1 cm² with a T80 lifetime of 800 h [7]. Similarly, an annealed Ti₃C₂T_x MXene interlayer forms triden-

tate Ti-O-P bonds with phosphonic acid heads on indium tin oxide (ITO), attaining a certified 26.44% PCE (champion 26.78%) on a 0.08 cm² area [71]. Bypassing substrate pre-treatments, introducing 4-dimethylaminopyridine (DMAP) as a solution catalyst accelerates phosphonic acid condensation via a transient phosphorylpyridinium intermediate, densifying the SAM to achieve a certified 26.44% PCE (champion 26.93%) with 95% retention after 700 h [72]. Where pinholes persist on nickel oxide (NiO_x), area-selective ALD of AlO_x passivates exposed oxide regions without coating SAM-covered domains, enabling a 64.68 cm² module PCE of 20.88% (champion 26.41%) [73], while treating NiO_x with cesium oxalate (CsOA) converts Ni³⁺ into [Ni(C₂O₄)₃]³⁻ and eliminates Ni⁴⁺ sites, supporting an 11.0 cm² device PCE of 26.48% with 85.7% retention after 1560 h at 65 °C [74].

Molecular shape has likewise been manipulated to regulate self-assembly thermodynamics. Incorporating dimethyl substituents onto spirofluorene cores imposes steric hindrance to force ordered molecular packing, yielding an uncertified champion PCE of 26.62% that maintains approximately 90% of its initial value after 1000 h of maximum power point tracking (MPPT) in nitrogen [75]. Bisphosphonate anchors in MeS-BPADCOP enforce vertical orientation through bidentate binding, reaching an uncertified 26.73% PCE with 94.2% retention after 1000 h under one-sun illumination [76]. A butterfly-shaped dibenzazepine core (PATDBAz) enables pi-pi stacking while preventing aggregation, sustaining 97.5% of initial efficiency after 1000 h under International Summit on Organic Photovoltaic Stability (ISOS) ISOS-L-1 testing (champion 26.47%) [77]. Intermolecular ordering can also be modulated by doping mixed SAMs to alleviate tensile interfacial strain, achieving a tandem champion PCE of 30.18% and an inverted 1.68 eV cell PCE of 23.48% (>90% retention after 1500 h) [78], or by post-assembling MeO-ADCOP onto pre-formed RS-2 monolayers to fill bare vacancies, achieving an uncertified 26.8% PCE with over 99% retention after 2000 h at 45 °C [79]. Despite these ordering schemes, monomeric SAMs exhibit weak shear resistance and detach under thermomechanical stress. Consequently, research shifted toward polymeric networks. Replacing small molecules with an asymmetric polycarbazole phosphonic acid (Poly-3Ph-4PACz) homogenizes flexible hole-selective contacts, yielding a certified tandem PCE of 32.95% (champion 26.13%, stabilised 27.12%) and a 57.6 cm² flexible module PCE of 22.22% [2]. Multidentate phosphonic acid polymer networks (P-SAM) increase interfacial fracture toughness nearly eightfold to 9.11 MPa (certified 26.61% on 0.06734 cm²) [80], while the phosphonic-acid-functionalized polymer PTPP combines continuous film formation with anchoring stability, sustaining 94.4% of initial performance after 800 thermal cycles (certified 26.4%, champion 27.1%) [81]. In tin-lead absorbers, blending hydrophobic trifluoromethyl molecules (MTFP) into poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) suppresses moisture uptake to reach 23.36% PCE (*V*_{oc} of 0.88 V) and a tandem PCE of 28.67% [82].

The third physical problem involves chemical coordination across the buried interface, which dictates heterogeneous nucleation, trap passivation, and ion migration. Buried contact chemistry directly regulates crystallization kinetics: imidazolium-functionalized SAMs guide crystal growth through pi-cooperative coordination with lead polyhalide complexes, retaining 93.8% of initial PCE after 936 h of operation at the maximum power point (champion 26.83%) [83]. On flexible substrates where poor precursor wetting impedes grain formation, modifying the SAM with NCGCl salt utilizes nitrile and guanidinium groups to promote heterogeneous nucleation and passivate defects, achieving a certified 26.64% PCE (champion 26.89%) [84]. Precursor pre-organization before coating also restricts phase competition: the zwitterionic additive CPTAC associates with formamidinium cations, contracting competing crystallization pathways to accelerate conversion into the photoactive alpha phase (uncertified 26.54% PCE on 0.1 cm²) [85].

Passivating deep defects at the buried interface has converged on chelating agents, heterojunction formation, and diffusion barriers. Inserting an aromatic amino acid derivative (NBP) between Me-4PACz and wide-bandgap absorbers coordinates unbonded Pb^{2+} cations and passivates iodide vacancies, enabling wide-bandgap single junctions to reach 20.81% PCE with a V_{oc} of 1.371 V and tandems to reach 29.05% [86]. Bifunctional cyanoacetic acid incorporated into both the SAM and perovskite precursor passivates interface boundaries, suppresses ion migration, and nearly doubles the reverse-bias breakdown voltage, achieving a tandem champion of 32.43% (stabilised 24.43%) [87]. In situ growth of gradient Ruddlesden-Popper two-dimensional/three-dimensional (2D/3D) heterostructures suppresses interface defects, reducing tandem current mismatch from 5.5% to 0.1% (certified 32.46%, champion 32.73%) [5], while conjugated TzTzI ligands construct 2D/3D networks that yield 24.5% on 1 cm^2 , 20.61% for 1.72 eV bandgaps, and 22.14% in mini-modules (champion 26.61%) [88]. Multi-site defect healing with thiazolidine derivatives (TZCl-CN) suppresses non-radiative recombination to reach an uncertified 26.39% PCE [89]. Bypassing HTLs entirely, bimolecular co-modification with THDPH and PEABr manages the perovskite/electrode boundary, retaining 90% of initial efficiency after 1000 h (champion 26.55%) [90]. To prevent lead leakage, polymerizing the crown ether DVB18C6 in situ yields a 95.14% Pb^{2+} sequestration efficiency (champion 26.78%) [91]. Ultrathin inorganic interlayers provide physical barriers to ion migration: ALD HfOx layers anchor 3F-PEAI via Hf-F interactions to block halide and silver diffusion (certified 26.6%, champion 27.1%) [92], paralleling SAM architectures that lose only 2% efficiency after 2100 h of MPPT at 65 °C under ISOS-L-2 conditions (certified 26.90%, champion 27.10%) [93].

Buried electron transport layers (ETLs) present analogous chemical and electrostatic challenges. Dual passivation combining DMDPTF with cobalt-acetate-modified tin oxide (SnO_2) produces an uncertified 26.41% PCE on 0.096 cm^2 and 22.18% on a 10.04 cm^2 module [94]. Reconstructing SnO_2 nanocrystals with in situ generated $[\text{Al}(\text{OH})_4]^-$ ions rebuilds the interfacial electrical double layer, delivering a certified 26.64% PCE (champion 26.7%) and 21.91% on a 65 cm^2 module [95]. For indoor applications, Mg-doping of SnO_2 widens the optical bandgap and reduces recombination to yield 35.54% under light-emitting diode (LED) illumination and 20.28% under one sun [96], while dual insulator contacts block shunting pathways to deliver an indoor PCE of 44.36% at 1000 lx and 43.08% at 600 lx (V_{oc} of 1.064 V, fill factor of 84.52%) [97].

5. A-Site and Halide Composition Engineering

Investigations into formamidinium lead triiodide (FAPbI_3) phase stabilization reach conflicting conclusions on how low-dimensional templates suppress the alpha-to-delta transition. Contact-triggered cationic interaction (CCI) asserts that structural refinement, cation homogenization, and defect suppression proceed via non-destructive physical contact between separately crystallized two-dimensional (2D) and three-dimensional (3D) perovskites without chemical mixing, interdiffusion, or permanent bonding (certified 25.61% power conversion efficiency (PCE), champion 26.25%) [98]. Conversely, formamidinium tetrafluoroborate (FABF_4) recrystallization demonstrates that constructing stable mixed-dimensional networks requires active formamidinium (FA^+)/methylammonium (MA^+) cation exchange and Ostwald ripening rather than passive contact, achieving an uncertified 21.36% champion PCE that retains 88% of initial efficiency after 10,000 bending cycles in nitrogen and 82% at $60\% \pm 5\%$ relative humidity (RH) [99]. This conflicts with crystallographic interface models where the alpha-to-delta transition nucleates epitaxially at a tensilely strained, coherent interface between alpha-phase (011) and delta-phase (110) planes, demanding 2D perovskitoid interfaces to disrupt delta-phase orientation (uncertified champion 25.61% PCE) [100]. Resolving whether stabilization

requires passive contact, active cation exchange, or orientation disruption demands operando cross-sectional depth profiling (such as cryogenic secondary ion mass spectrometry) paired with transmission electron microscopy under thermal bias to establish whether bulky A-site cations stay segregated or interdiffuse across the interface.

Precursor stabilization strategies over the past six months divide over whether failure originates from proton transfer or physical volatility. Competitive proton-transfer suppression using methyl trifluoroacetate prevents FA+ deprotonation to extend precursor shelf life beyond 22 days, producing uncertified champion devices reaching 26.35% PCE that retain 94.75% efficiency after 1464 h of humidity exposure [101]. In single crystals, citric acid similarly mitigates cation deprotonation, scavenges oxidized iodine (I₂/I₃⁻), and passivates lead (Pb²⁺) dangling bonds to yield 23.39% PCE [102], while multidentate ZL1 anchors suppress volatile FA loss to lift PCE from 24.20% to 26.13% [103]. In contrast, studies on volatile additives demonstrate that cation volatility dictates structural evolution: in thiocyanate systems, ammonium (NH₄⁺) evaporates completely during annealing to drive bulk crystallization, whereas non-volatile phenylethylammonium (PEA⁺) segregates to grain boundaries for interfacial passivation (23.3% PCE) [104]. In cesium lead triiodide (CsPbI₃), formamidine acetate (FAAC) intermediate engineering lowers Cs₄PbI₆ formation energy, where transient FA+ intercalation into DMAPbI₃ accelerates dimethylammonium (DMA⁺)/cesium (Cs⁺) exchange and dimethylammonium iodide (DMAI) volatilization (21.84% PCE) [105]. Conversely, ethyl thioacetate post-treatment bypasses cation exchange to passivate uncoordinated Pb²⁺ through Lewis base coordination (17.07% PCE; T80 of 1360 h) [106]. Rubidium chloride (RbCl) guides phase transformations in printable mesoscopic methylammonium lead triiodide (MAPbI₃), though performance is constrained to 18.48% PCE with only 300 s of steady-state tracking [107]. Machine learning via LEAP couples large language models with active learning to prioritize 6-CDQ and 2-CNA additives, lifting average PCE from a 19.25% control to 20.13% and 20.87% (21.32% champion) [108], while vacuum-assisted urea coordination modulates triple-cation crystallization to reach 19.61% PCE [109].

Lattice microstrain from A-site mismatch divides stabilization strategies between static coordination and dynamic mechanical regulation. Static passivators such as guanidinium methanesulfonate (GUAMS) bridge FA+ via hydrogen bonding while passivating Pb²⁺, achieving 26.11% PCE in inverted cells, 24.49% on flexible substrates, and a T80 of 1200 h [110]. Zwitterionic Cl-PPS additives identically coordinate Pb²⁺ and hydrogen-bond with FA+ to reach 26.61% PCE [111]. In n-i-p devices, thiacetazone yields 26.52% PCE in ambient air and 23.24% at 70% RH [112], while fluoropyridine additives (DFPy) suppress non-radiative recombination to reach 26% PCE [113]. Because static bonds cannot accommodate cyclic thermal expansion, dynamic strain regulation via thermal disulfide ring-opening crosslinks grain boundaries, maintaining 95.7% of initial efficiency after 1,800 h under maximum power point tracking (MPPT) with day-night cycling (26.5% champion PCE) [114]. Fluorinated misplaced-dipole elastomers similarly suppress thermal stress cracks, yielding 25.54% PCE on flexible devices versus 26.83% on rigid substrates [115]. At module scale, out-of-plane strain homogenization through A-site doping and an in situ 2D stress-relief layer delivers a certified 25.55% PCE (26.59% champion) across 100 cm² under International Summit on Organic Photovoltaic Stability (ISOS)-L-1 testing [13]. Solvent-free dry vacuum deposition independently demonstrates that minimizing microstrain yields a standard deviation of 0.36% across 90 cells (25.4% champion PCE) [116].

In wide-bandgap (WBG) mixed-halide perovskites, light-induced halide segregation divides mechanistic interpretations between differential crystallization kinetics and lattice microstrain. Eutectic urea

phosphate demonstrates that halide demixing stems from unequal growth kinetics; stronger affinity for bromine-based over iodine-based octahedra synchronizes crystallization, eliminating vertical and lateral phase segregation (22.27% PCE) [117]. TTA additives corroborate this kinetic synchronization mechanism, achieving 20.55% PCE and an open-circuit voltage (Voc) of 1.339 V [118]. In contrast, methylammonium-guanidinium (MA-GA) codoping demonstrates that balancing lattice strain directly stabilizes the lattice against halide demixing without kinetic agents, retaining over 95% of initial efficiency after 1400 h of continuous operation in 1.77 eV cells (21.5% PCE) [119]. Halide substitution also dictates transport limits in thick absorbers: in Cs_{0.05}(FA_{0.90}MA_{0.10})_{0.95}Pb(I_{0.95}Br_{0.05})₃ middle cells, performance drops above 450 nm, but chloride substitution permits thickness scaling up to ~700 nm without sacrificing fill factor (FF) or Voc (23.4% PCE on 1 cm²) [120]. Functional group-modulated benzoic additives suppress non-radiative recombination in WBG single junctions to exceed 22.9% PCE, enabling 2-terminal tandem cells surpassing 31% [121]. Multifunctional ATAA molecules stabilize both FA⁺/MA⁺ cations and I⁻/Br⁻ halides via hydrogen bonding, upshifting the valence band maximum (VBM) by 0.12 eV (25.82% PCE on 1 cm²) [122], while dual A-site 1D (PDA)_{0.6}(GUA)_{1.4}Pb₆I₁₆ bridges suppress iodide migration and interfacial recombination, attaining 20.46% PCE (Voc 1.328 V) in WBG cells and 28.43% in all-perovskite tandems with a T80 of 1000 h [123].

Low-dimensional heterojunctions introduce fundamental trade-offs between surface passivation and transport resistance. Precursor incorporation of long-chain oleylammonium ligands forms in situ 2D/3D heterojunctions yielding 26.22% PCE on small areas, 24.6% on 1 cm², and 21.1% on 6.8 cm² mini-modules (T80 of 1050 h) [124]. Non-Dion-Jacobson (non-DJ) 2D perovskites using 3,3'-methylenediphenyldiammonium cations utilize organic bilayer pi-pi interactions to achieve 26.52% PCE on 50 cm² modules [125]. At single-crystal interfaces, layered 1D (4PPY)PbI₃ blocking layers suppress iodide migration more effectively than isolated 1D or 2D phases, retaining 100% of initial efficiency after 1032 h continuous illumination (25.4% PCE) [126]. Alkyl chain optimization with tetrahexylammonium iodide (THAI) generates 1D THAPbI₃ heterojunctions that anchor FA⁺ via hydrogen bonding, suppressing outward cation diffusion and strain gradients (25.06% PCE) [127]. Grain-boundary encapsulation via 4-FAPyCl₂ in PbI₂ precursors forms quasi-2D coatings that protect 3D alpha-FAPbI₃, limiting ambient degradation to less than 3% PCE loss (25.32% PCE) [128]. Precursor addition of 0.3% MeO-PEACl raises Voc from 1.11 to 1.16 V and PCE from 21.1% to 23.1%, maintaining the black phase after 3380 h at 65% ± 10% RH [129]. To overcome spacer insulation, thiophene-2-hydrazide (ThCH) induces interlayer orbital coupling in 2D Ruddlesden-Popper (RP) phases, though its certified 21.74% efficiency (22.41% champion) was evaluated on a diminutive 0.074 cm² aperture [130]. Pre-formed n=1 2D nanosheets seed 3D oriented growth to reach 21.36% PCE (20.75% stabilized output) with hysteresis below 1% [131], and poly(3-hexylthiophene) (P3HT)-modified 2D quantum dots suppress interfacial defects to reach 25.04% PCE [132]. Isomeric cytosine additives outperform iso-cytosine through delocalized bonding in quasi-2D films (22.4% PCE; T80 of 3600 h) [133], and propylammonium iodide in two-step blade coating directs (111) grain orientation in inverted cells (20.05% PCE; 95% retention after 1680 h MPPT) [134]. Conversely, unoptimized capping layers such as 2D (4-FBA)2PbBr₄ yield an uncertified 18.4% PCE and retain only 77.82% after 10 h at 80 °C and 55.32% after 30 days in ambient air [135], while vacuum co-evaporated PEA-passivated films reach 17.79% PCE [136].

6. Tandem Architecture and Current Matching

Investigations into monolithic two-terminal (2T) perovskite/silicon tandems diverge on whether a laterally conductive transparent conducting oxide (TCO) interlayer is required for subcell recombination. Conformal coverage on textured silicon heterojunction (SHJ) bottom cells reportedly required an 8 nm indium tin oxide (ITO) layer modified with nickel oxide (NiO_x) and a polymeric self-assembled monolayer (Poly-SAM), gaining 0.85 mA cm^{-2} in short-circuit current density (JSC) to reach 20.82 mA cm^{-2} in a 31.8% champion cell [137]. Conversely, omitting indium-based oxides entirely, a reactive plasma deposition (RPD) tin oxide (SnO_x) recombination layer directly anchored self-assembled monolayers (SAMs) to suppress halide migration and nonradiative recombination, achieving certified 33.6% power conversion efficiency (PCE) (33.2% champion on 1 cm^2) and certified 31.0% on a 207.9 cm^2 minimodule [1]. The physical controversy centers on whether removing ITO lateral conductivity creates localized resistive bottlenecks across textured pyramids. Spatially resolved voltage-dependent photoluminescence (PL) imaging via the local transport and recombination study (LTRS) method settles this debate: by mapping series resistance (R_s) to isolate an absolute 1.9% efficiency loss in a 29% monolithic tandem while decoupling current mismatch, low shunt resistance, and luminescence coupling in finite element simulations, operando LTRS determines whether TCO-free junctions exacerbate resistive losses under spectral variation [138].

Parasitic optical absorption within selective contacts further constrains current matching. Thinning electron-selective fullerene (C_{60}) contacts to 1 nm yields thickness-independent extraction, boosting tandem JSC by 0.24 mA cm^{-2} in a 31.7% champion cell [139]. Yet atomic layer deposition (ALD) of tin oxide damages underlying perovskite during C_{60} capping; substituting ALD with thermally evaporated antimony oxide eliminates damage while permitting thinner fullerenes, improving top-cell JSC by $\sim 1 \text{ mA cm}^{-2}$ (certified 27.7%, champion 28.16% on 64.64 cm^2) [140]. In normal (n-i-p) tandems, an ultrathin 10 nm dopant-free 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (spiro-OMeTAD) layer reduces parasitic absorption by 92.2% and reflection losses by 18.4%, delivering an uncertified 29.73% PCE on 0.135 cm^2 and 28.77% (certified 28.25%) on 1.012 cm^2 [141]. Decoupling current matching, four-terminal (4T) architectures using semitransparent 1.67 eV wide-bandgap (WBG) subcells with vanadium oxide-nickel oxide ($\text{VO}_x\text{-NiO}_x$) contacts achieved 21.68% on 0.148 cm^2 , retaining 98% efficiency after 1472 h maximum power point tracking (MPPT) at 25°C (31.25% tandem on 1 cm^2) [142], while priming CsPbX_3 intermediate phases yielded 20.91% on 0.1 cm^2 and 19.67% on 1.0 cm^2 (31.04% tandem on 1.0 cm^2) [143].

Matching bottom cells requires WBG absorbers (1.67-1.85 eV), where drift-diffusion modeling indicates that bromide substitution and cesium alloying in formamidinium lead triiodide (FAPbI_3) increase carrier effective mass from 0.173 me to 0.243-0.258 me, restricting transport despite projected 32.10% PCE and 2.05 V open-circuit voltage (VOC) [144]. Approaches over the six-month period diverge on additive chemistry. Several groups introduced fluorinated agents: difluoro(oxalato)borate and tetrafluoroborate synchronized crystallization without intermediate halide redistribution, certifying a 30.3% tandem [145]; volatile octafluoronaphthalene (OFNA) relieved tensile stress and passivated defects (22.23% single junction, 31.43% tandem) [146]; and 4-(trifluoromethyl)benzenesulfonyl chloride (BSCF_3) coordinated lead and formamidinium (19.81% single junction, 1.32 V VOC, 27.26% tandem) [147]. In direct conflict, tracking ion diffusion revealed that mobile fluoride from lithium fluoride (LiF) interlayers penetrates the perovskite bulk, accelerating halide segregation and degrading photostability; an ion-blocking lithium-theophylline (Li-T) interlayer was required to halt fluoride migration, securing 1.284 V VOC and 19.55% efficiency in 1.77 eV cells (28.41% tandem) [148].

Non-fluorinated strategies instead employed forchlorfenuron (CPPU) to direct (100) facets (certified 23.46% single junction, 1.295 V VOC; certified 31.93%, champion 32.17% tandem on 1.0 cm²) [3], 1,2-diiodotetrafluorobenzene (1,2-DITFB) to retard bromide crystallization (20.25% single junction, 1.34 V VOC, 28.58% tandem) [149], phenylhydrazine derivatives to inhibit iodide oxidation (20.75% single junction, 28.63% tandem) [150], and 2-phenylethylammonium chloride (PEACl) modification with solvent etching (CDSE) to remove residual lead iodide (PbI₂) (21.77% single junction, 31.85% tandem) [151]. Rubidium cyanate (RbOCN) lattice incorporation stabilized 1.73 eV absorbers (22.45% single junction, 25.37% on 1 cm², 26.75% champion on an uncertified 0.09 cm² area, 2.22 V VOC) [152], while targeted passivator coordination achieved a certified 2.00 V tandem VOC (certified 31.52%, champion 32.74%) [153]. In contrast, additive-free stoichiometry control via a PbI₂-to-total PbX₂ ratio of ~55-75% slowed crystallization to deliver single-junction PCEs of ~24.19% (1.67 eV) to 16.83% (1.93 eV) and a 29.1% tandem [154]. Evaporated scaffolds expose a mechanistic disconnect: two-step thermal evaporation yielded stabilized tandem PCEs above 29% [155], yet five sequentially evaporated scaffold stacks produced identical ~19% single-junction PCEs (28.4% champion tandem) despite profound variations in precursor conversion and morphology [156].

At buried hole-selective junctions, composite SAMs of 4,4',4'',4'''-methanetetrayltetrabenzoic acid (4MA) with [4-(3,6-dimethyl-9H-carbazol-9-yl)butyl]phosphonic acid (Me-4PACz) on NiO_x use tetrahedral molecular pinning to passivate defects, achieving 23.52% in 1.68 eV cells and a certified 32.1% (32.26% champion on 1 cm²) tandem [12]. Aligning packing via intermolecular pi-pi interactions in [2-(3,6-dimethyl-9H-carbazol-9-yl)ethyl]diphenylphosphine oxide (Me-Ph2mPACz) reduced non-radiative losses from 168 mV to 124 mV (single-junction PCE 21.82% to 23.14%), delivering a certified 32.45% (33.4% champion) tandem retaining 83% efficiency after 1000 h MPPT at 85 °C in air [8]. Three-dimensional C-H...pi transport networks from asymmetric MeO-BADCz SAMs certified at 33.04% (24.05% champion reported) [11], and the triazole-based quaternary ammonium molecule TMTEA co-regulated self-assembled monolayer (SAM) packing and orientation (23.93% single junction, 32.15% champion) [157]. Under operational current mismatch, subcells are driven into reverse bias. In all-perovskite tandems combining WBG and narrow-bandgap (NBG) absorbers, sustained reverse bias induces a destructive redox cascade: hole injection oxidizes iodide to iodine species that oxidize Sn²⁺ to Sn⁴⁺, driving lattice collapse in a 29.03% tandem [158].

7. Degradation Mechanisms Under Operation

Studies of operational degradation reveal an unresolved mechanistic contradiction regarding whether bulky molecular passivators relieve interfacial lattice strain or accelerate degradation through local thermal expansion. Bulky passivators are widely applied to terminate dangling bonds and impede defect formation, with molecular locks such as 3-PMPCl raising the phase transition barrier in formamidinium lead triiodide (FAPbI₃) by suppressing [PbI₆] octahedral disorder to reach a certified 27.6% power conversion efficiency (PCE) [159]. Similarly, facet-engineering additives such as sulfonates relieve residual lattice strain by redirecting crystallization kinetics [160], and molecular buffering layers prevent mechanochemical fatigue under thermal cycling [161]. However, direct operando structural analysis demonstrates that bulky anionic passivators induce local lattice expansion under thermal stress at heterointerfaces, driving interfacial fatigue even while unencapsulated devices retain 87% of their initial efficiency after 4092 h of maximum power point tracking (MPPT) under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-1 with a time to 80% of initial performance (T80) of 3566 h [162]. This finding directly challenges the assumption that bulky organic modifiers serve as inert physical anchors. The contradiction cannot

be resolved by post-mortem spectroscopy because lattice relaxation occurs upon cooling; settling it requires in situ, temperature-dependent grazing-incidence wide-angle X-ray scattering paired with depth-resolved photoelectron spectroscopy under operational bias to quantify local lattice expansion directly against defect formation rates.

A parallel divergence divides studies of buried hole-selective interfaces in inverted (p-i-n) architectures. One body of literature identifies out-of-plane molecular desorption from metal oxides as the primary degradation mode, which can be mitigated by enforcing spatial confinement with a rigid planar backbone achieving 3.72 Å pi-pi stacking [163] or by synthesizing polymeric self-assembled monolayers (SAMs) such as Poly-4PADCB to reduce steric hindrance [164]. Conversely, competing evidence attributes contact failure to spontaneous molecular aggregation that leaves bare oxide domains, an issue countered through asymmetric molecules with steric hindrance like PhBr-4PACz that retain over 96% efficiency after 2000 h at 65 °C under ISOS-L-2 [165] or fluorinated confinement interlayers [166]. A third perspective indicates that self-assembled monolayer (SAM) degradation occurs predominantly during perovskite solution processing via solvent stripping, requiring transient 2D perovskite sacrificial barriers [167] or hydroxyl enrichment using MXene additives [168]. Although amino acid hydrochloride additives [169] and tailored hole transport materials like WH13 [170] yield stability claims, these evaluations frequently rely on tiny active areas (such as 0.058 cm² in [169]), where edge encapsulation masks real-world planar degradation. In perovskite/silicon tandem cells, co-assembled siloxane monolayers [10] and amorphous asymmetric contacts [6] retain 87.12% and 87% initial PCE after 1000 h of MPPT at 85 °C, respectively, confirming that resilient chemisorption retards thermally activated interfacial delamination.

In the perovskite absorber, the consensus that device decay is governed primarily by bulk mobile ions has been contested by direct quantification of deep-level defect kinetics. Operational degradation is dominated by deep-level interstitial and antisite defects, specifically iodine on formamidinium sites (I_FA) and lead sites (I_Pb), which drive performance decay despite existing at concentrations three orders of magnitude below shallow defects [171]. Passivating these deep traps with 3TU2+ reduces rear-interface energy losses from 1.46% to 0.62% [171]. In contrast, widespread reports continue to focus on macroscopic ionic flux, using liquid additives such as 1-BP to suppress ion migration as tracked by time-of-flight secondary ion mass spectrometry (ToF-SIMS) [172], positional methylation isomers like 5-MH that retain 92% efficiency after 2000 h under ISOS-L2 [173], or eperisone hydrochloride to eliminate residual dimethyl sulfoxide (DMSO) and maintain 90.39% efficiency after 1200 h under ISOS-L-3 [174]. Other interventions combat defect strain through trifluoroethylammonium chloride (3TFACl) additives [175] or 1D (PCOZ)PbI₃ interlayers that preserve 94.2% efficiency after 1000 h under ISOS-L-3 [176].

Operational stability is also restricted by charge transport layers, where conflicting failure mechanisms are assigned to identical materials. In inverted devices, phenyl-C61-butyric acid methyl ester (PCBM) degrades via photo-induced [2+2] cycloaddition dimerization under illumination, inducing aggregation and morphological decay that nonplanar tetraphenylethylene additives prevent [177]. Yet separate studies treat fullerene decline as an interfacial corrosion problem solved by buffer layers like BAE-Phen that coordinate migrant metal ions to retain 90.5% PCE after 2000 h at 85 °C [178], solvent-free multicomponent fullerene composites [179], or bridging interlayers tested on 0.045 cm² cells with a T80 of 500 h [180]. In normal (n-i-p) devices employing 2,2',7,7'-tetrakis[N,N-di(4-methoxyphenyl)amino]-9,9'-spirobifluorene (Spiro-OMeTAD), a fundamental trade-off persists: promoting lithium bis(trifluoromethanesulfonyl)imide (LiTFSI) ion-pair dissociation is necessary to

sustain electrical conductivity [181], yet uncoordinated lithium ions migrate rapidly into the absorber. Suppressing this degradation channel requires either phenothiazine ionic anchors to bind lithium, yielding 90% retention after 800 h of MPPT on 0.04 cm^2 cells [182], or lithium-based deep eutectic matrices that deliver 90.5% retention after 1800 h under ISOS-L-1I [183].

These mechanistic debates unfold against an acute reporting deficit across the field: of 872 closely read papers published over the half year, only 39 reported a T80 value and only 39 specified an ISOS testing protocol. Although champion operational lifespans reached 4100 h [184], 4000 h [185], 3600 h [133], and 3566 h [162], the rarity of standardized aging procedures leaves the relative severity of interfacial expansion, ion accumulation, and transport layer degradation obscured by divergent testing metrics.

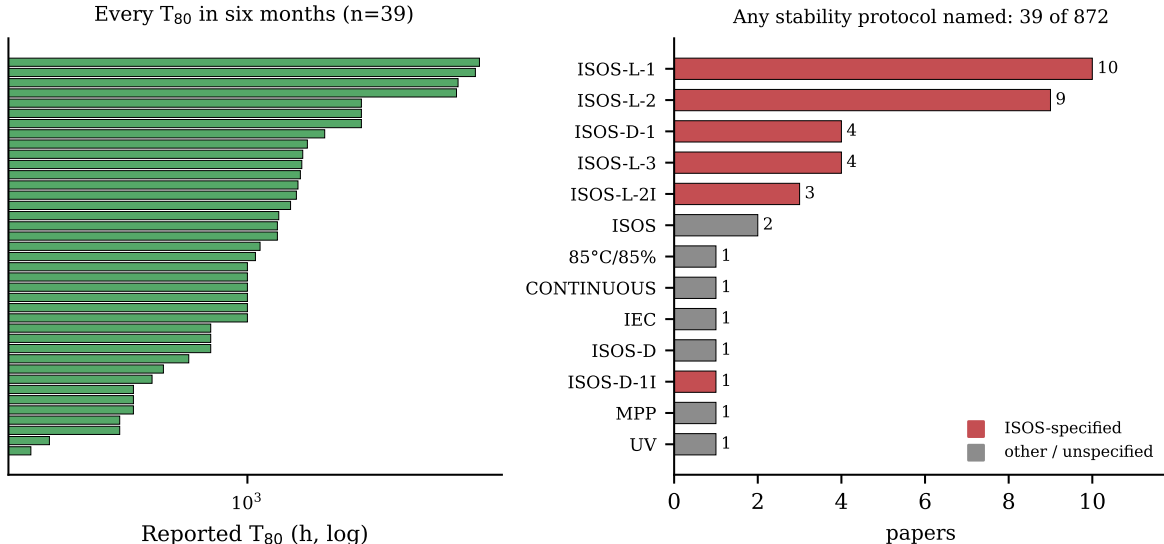


Figure 2. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

8. Large-Area Coating and the Area Penalty

Conflicting kinetic models across recent scalable coating studies reflect an unresolved physical disagreement over whether precursor-solvent coordination accelerates or impedes uniform perovskite crystallization. One framework asserts that prolonged solvent-precursor interaction time during blade coating stabilizes detrimental solvent-adduct intermediate phases that degrade film crystallinity, demanding rapid adduct suppression [186]. In direct contradiction, alternative investigations conclude that strong coordination by N-methyl-2-pyrrolidone (NMP) in dimethylformamide (DMF)-free 2-methoxyethanol (2-ME) or isopropanol (IPA) is necessary to stabilize intermediate complexes and delay premature surface nucleation, thereby increasing open-circuit voltage (Voc) from 1.08 to 1.09 V and fill factor (FF) from 75% to 80.5% [187,188]. This kinetic divergence extends across solvent systems: ternary dimethyl sulfoxide (DMSO), acetonitrile (ACN), and 1,3-dimethyl-2-imidazolidinone (DMI) mixtures rely on volatile ACN to accelerate initial nucleation during slot-die printing to reach 20.62% power conversion efficiency (PCE) [189], whereas reduced-temperature bar coating at 100 °C employs low translation speeds of 0.3 mm/s to sustain a metastable supersaturation zone rather than accelerating precipitation [190]. Resolving whether solvent adducts degrade or preserve large-

area films requires operando spatial diagnostics; coupling in situ photoluminescence (PL) and diffuse reflectance during quenching with neural networks captures non-uniform crystallization before localized shunts solidify [191,192]. Precursor colloid engineering via formate anions restricts cluster size to around 30 nm [193], and polypropylene glycol (PPG) aids formamidinium lead iodide (FAPbI₃) microcrystal synthesis [194]. However, claims of defect suppression through adduct control remain restricted to uncertified, sub-0.1 cm² cells (such as 26.5% PCE on 0.0665 cm² [186] or certified 26.22% on 0.06 cm² [194]), without proving that intermediate coordination equilibria survive roll-to-roll drying gradients under mechanical shear [195].

A second major source of the area penalty originates in charge transport layer (CTL) non-uniformity and buried contact resistance across large apertures. For electron transport layers (ETLs), atmospheric-pressure spatial atomic layer deposition (AP-SALD) of tin oxide (SnO_x) achieves 20% PCE following 180 °C annealing, yet conformal deposition replicates substrate roughness, which impairs perovskite crystallization [196]. Solution approaches attempt to mitigate interfacial roughness: electrostatic regulation of SnO₂ nanoparticle assembly yields uniform coatings, delivering 24.70% PCE at 1 cm² and over 21% on a 65.24 cm² module (certified 26.03% on 0.0727 cm²) [197], while multicycle infrared annealing of SnO₂ quantum dot (QD) ETLs limits cracking, reaching 17.09% at 1 cm² and 15.18% at 2 cm² compared to 12.78% for conventional hotplate annealing [198]. For hole-selective contacts, self-assembled monolayers (SAMs) suffer severe dewetting during large-area coating; incorporating 3-BPIC-F into MeO-4PACz enhances dipole alignment and suppresses pinholes, raising module PCE from 16.43% to 19.80% (champion 23.64%) [199], while low-polarity alkane and alcohol solvents enable blade coating of SAMs at 80% relative humidity (RH) to yield a certified 26.1% PCE [200]. At the counter interface, non-uniform chemical doping induces spatial conductivity variances, which ferrocenium radical cation dopants in Spiro-OMeTAD eliminate to achieve 26.13% in cells and 22.21% in modules [201].

Hydrodynamic instability and resistive interconnect losses govern the final translation from single devices to large modules. Drying flows induce severe thickness gradients; fluid-locking interfacial layers counteract coffee-ring segregation to yield 21.39% PCE on a 100 cm² module [202], whereas slot-die coated FAPbI₃ without fluid control drops from 20.8% on small areas to 19.1% at 8.4 cm² and 14.8% at 59.5 cm² active area [203]. Post-deposition passivation attempts to seal boundary defects: immersing hot-bladed films into cold strontium iodide (SrI₂) isopropanol solutions produces 20.32% on a 10.13 cm² module [204], and pneumatic spray coating forms 1D PyPbI₃ surface barriers (champion 21.64%) [205]. Fully dry vacuum co-evaporation avoids wet-film fluid mechanics entirely, using PbCl₂ additions to disrupt PbI₂ stacking and reach over 38% efficiency at 900 lux on a 3.9 cm² indoor module [206]. Beyond 100 cm², mechanical deformation and interconnect resistance dominate: dynamic covalent cross-linking monomers embedded in perovskites provide self-curing at 40 °C for 30 min, yielding a certified 26.8% and champion 27.12% PCE across 655.2 cm² flexible modules [15], while liquid metal interconnects maintain continuity over a 900 cm² platform [18].

9. Cross-Cutting Synthesis: What the Mechanisms Say Together

A fundamental disagreement across recent multi-junction investigations concerns whether non-radiative voltage deficits originate from defective charge extraction at buried contacts or from trap-assisted recombination within bulk wide-bandgap (WBG) absorbers. In monolithic two-terminal (2T) perovskite/silicon tandems, one framework attributes performance bottlenecks primarily to hole-extraction resistance and halide accumulation at the buried contact, demonstrat-

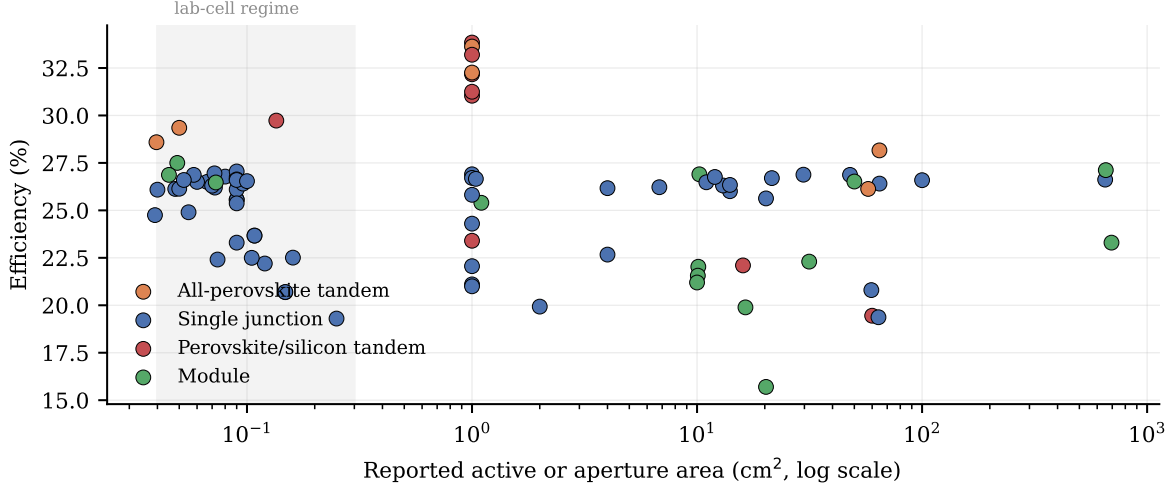


Figure 3. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

ing that replacing indium tin oxide (ITO) with a reactive plasma deposition (RPD) tin oxide (SnOx) recombination layer directly anchored to self-assembled monolayers (SAMs) delivers a certified power conversion efficiency (PCE) of 33.6% (champion 33.2% on 1 cm²) and a certified 31.0% on a 207.9 cm² minimodule [1]. Conversely, competing investigations conclude that carrier losses are dominated by photoinduced phase segregation and deep-level defect generation in the bulk WBG film, asserting that replacing monomeric molecules with polymeric polycarbazole phosphonic acid (Poly-3Ph-4PACz) relieves bulk-driven recombination to reach a certified 2T tandem PCE of 32.95% (champion 26.13%, stabilised 27.12% in single junctions; 22.22% on a 57.6 cm² flexible module) [2]. Standard steady-state photoluminescence (PL) and current density-voltage characterization cannot decouple these hypotheses because contact recombination velocity and bulk quasi-Fermi level splitting collapse produce identical terminal open-circuit voltage (Voc) losses. Deciding whether the primary voltage deficit resides at the contact or in the absorber requires operando, spatially resolved voltage-dependent photoluminescence imaging via the local transport and recombination study (LTRS) method coupled with bias-dependent transient absorption spectroscopy. By mapping series resistance to isolate a 1.9% efficiency loss in a 29% monolithic tandem while decoupling current mismatch, low shunt resistance, and luminescence coupling in finite-element simulations, operando LTRS determines whether carrier losses originate from interface extraction barriers or bulk non-radiative recombination [138].

Reading the separate axes together reveals that performance gains claimed in one domain routinely impose physical penalties in another. At the perovskite top surface, constructing low-dimensional capping overlayers or deploying bulky organic ammonium passivators terminates dangling bonds and heals halide vacancies, achieving certified PCE values of 26.2% (champion 26.49%) with a two-dimensional/three-dimensional (2D/3D) guanidinium heterostructure [20], 26.15% (champion 26.71% on 1 cm²) through one-dimensional (1D) TEAPbI₃ phase reconstruction [19], and a certified 27.6% via 3-PMPCl molecular locking [159]. However, evaluating these capping layers against operational degradation demonstrates an incompatible physical demand: bulky organic cations introduce anisotropic out-of-plane transport barriers that trap photogenerated carriers under operating electric fields, while operando structural measurements reveal that bulky passivators undergo pronounced local thermal

expansion relative to the perovskite lattice. Under thermal stress, this expansion generates localized microstrain and interfacial fatigue, driving degradation even in unencapsulated devices that retain 87% of initial efficiency after 4092 h of maximum power point tracking (MPPT) under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-1 with a time to 80% of initial performance (T80) of 3566 h [162]. Incorporating facet-engineering additives [160] or molecular buffering interlayers [161] to relieve residual lattice strain demonstrates that chemical defect passivation cannot be decoupled from thermomechanical fatigue: chemical saturation of surface coordination bonds trades immediate electronic gains for latent operational instability under thermal cycling.

An analogous conflict governs the buried interface, where electrostatic optimization clashes with mechanical adhesion and lateral conductivity. Functionalizing carbazole phosphonic acid SAMs with polar cyanoacetic acid co-adsorbents nearly doubles the reverse-bias breakdown voltage to yield an uncertified tandem champion PCE of 32.43% (stabilised 24.43%) [87], while asymmetric Br-4PhCz monolayers deliver a certified tandem PCE of 28.98% (champion 29.35%) on 0.05 cm² [63]. Yet monomeric SAMs exhibit low fracture resistance and undergo chemical oxidation and mechanical delamination under operational stress. Polymeric crosslinking mitigates interfacial shear failure, as multidentate polymer networks (P-SAM) increase interfacial fracture toughness nearly eightfold to 9.11 MPa (certified 26.61% on 0.06734 cm²) [80] and phosphonic-acid-functionalized polymers (PTPP) sustain 94.4% of initial efficiency after 800 thermal cycles (certified 26.4%, champion 27.1%) [81]. Nevertheless, crosslinked polymeric interlayers create lateral transport resistance across textured silicon bottom cells, where conformal coverage on random pyramids requires either an 8 nm ITO interlayer with nickel oxide and polymeric SAMs (gaining 0.85 mA cm⁻² in short-circuit current density to reach 20.82 mA cm⁻² in a 31.8% champion tandem) [137] or direct self-assembled monolayer (SAM) anchoring on SnOx [1]. Contact design therefore cannot optimize dipole energetics, mechanical adhesion, and lateral charge extraction simultaneously.

Finally, precursor coordination kinetics reflect an unbridged disconnect between sub-0.1 cm² efficiency records and scalable manufacturing. While non-destructive contact-triggered cationic interaction (CCI) yields a certified 25.61% PCE (champion 26.25%) [98] and microcrystal growth with polypropylene glycol achieves a certified 26.22% on 0.06 cm² [194], solvent-coordination models reach diametrically opposing conclusions. One framework demands rapid suppression of intermediate solvent adducts to avoid structural defects (uncertified 26.5% on 0.0665 cm²) [186], whereas another insists that strong coordination by N-methyl-2-pyrrolidone is required to delay premature nucleation, increasing Voc from 1.08 to 1.09 V and fill factor (FF) from 75% to 80.5% [187]. These chemical mechanisms do not converge because the intermediate equilibria that suppress defect formation in static, small-area spin coating fail under the mechanical shear and drying gradients of scalable deposition [195]. Rather than converging on a shared physical mechanism, the literature remains bifurcated between small-area electrostatic optimization and the thermo-chemo-mechanical realities of operational modules.

10. Reporting Practice Over Six Months

A fundamental disagreement in inverted perovskite solar cells (PSCs) concerns how self-assembled monolayer (SAM) hole-selective contacts degrade during operation: while one investigation attributes operational failure to out-of-plane molecular desorption from metal oxides that rigid planar conjugation suppresses [163], another concludes that degradation proceeds via field-induced lateral aggregation into isolated clusters [165]. Resolving whether molecular detachment or lateral dewetting governs

contact failure requires operando bias-dependent sum-frequency generation spectroscopy paired with in situ conductive atomic force microscopy under continuous maximum power point tracking (MPPT) at operational temperatures. Reconciling such conflicting degradation mechanisms across the literature is hindered by the scarcity of standardized testing protocols: when stability assessments rely on disparate stress regimes, distinguishing true interfacial stabilization from divergent testing conditions becomes impossible.

Across the six months, pooled reporting rates over 2238 indexed works reveal substantial deficits in primary performance and reliability metrics. Only 52.1% (1167 of 2238) of abstracts stated any power conversion efficiency (PCE) value. Independent verification was rare: only 8.5% (191 of 2238) reported an independently certified efficiency, leaving most performance claims susceptible to unverified spectral mismatch or contact overestimation. Stabilised performance tracking appeared in only 11.0% (246 of 2238) of abstracts, while reporting of current-voltage hysteresis was confined to 2.7% (60 of 2238), allowing capacitive transients and ion accumulation to obscure steady-state power output. Device geometry was similarly omitted, with only 8.4% (188 of 2238) stating the active area. Standardized operational aging was almost entirely absent, with only 1.7% (38 of 2238) identifying an International Summit on Organic Photovoltaic Stability (ISOS) protocol label. For a reader attempting cross-device comparisons, these low detection rates mean that nominally equivalent efficiencies often represent fundamentally incomparable physical measurements. These detection rates reflect automated parsing of published abstracts rather than an exhaustive audit of experimental methodologies; omitting a testing protocol or geometric parameter in an abstract does not establish that the underlying investigation failed to perform the measurement.

Examining monthly ranges demonstrates that reporting practices did not improve across the half year. Stating an efficiency sat flat within a narrow monthly range of 48.1% to 55.3%. Reporting of certified efficiencies oscillated between 5.9% and 11.6%, while stabilised efficiency remained confined to a tight monthly range of 9.2% to 13.5%. Active area disclosure sat flat between 5.8% and 10.3%, hysteresis identification stayed low within 1.5% to 4.3%, and ISOS labeling remained marginal between 0.6% and 2.7%. Although the monthly share of closely read papers expanded steadily from January (12.6%), February (15.0%), and March (16.7%) through April (18.8%), May (17.7%), and June (19.2%), this growing literature volume produced no upward drift in reporting completeness. Practice did not evolve; instead, reporting metrics occupied fixed corridors throughout the entire six months.

Exemplary reporting occurs where full geometric, certification, and operational parameters converge. Large-area module studies provide verified baselines by simultaneously stating aperture dimensions, certified efficiencies, and standardized degradation conditions, including an active area of 100 cm² with a certified PCE of 25.55% under ISOS-L-1 tracking [13] and 100 cm² with a certified PCE of 23.55% under ISOS-D-1 storage [14], establishing reproducible benchmarks for scalable interconnects [13,14]. Standardized operational testing is also illustrated by small cells identifying an ISOS-L-2 protocol and certified PCE of 21.54% alongside a time to 80% of initial performance (T80) of 1210 h [207], continuous operational stress under ISOS-L-3 [174], or ISOS-L2 testing with a certified 26.78% PCE [173], providing unambiguous tracking of photothermal decay [173,174,207]. In sharp contrast, absent reporting obscures physical mechanisms. Extreme claims, such as an uncertified champion PCE of 28.49% [208], omit certification, active area, and standardized aging protocols from the abstract. High champion efficiencies are frequently reported on sub-0.1 cm² cells, such as 0.072 cm² at 26.2% PCE [209] or 0.058 cm² at 26.87% PCE [169], where lateral current collection and edge encapsulation distort planar transport physics [169,209]. At the opposite extreme, scaling

studies describe an aperture of 900 cm^2 without providing an independently certified efficiency [18]. Uncertified claims evaluated on tiny cells without standardized protocols impede physical comparison and leave operational longevity unanchored.

11. Research Gaps and Outlook

A fundamental disagreement in multi-junction investigations concerns whether non-radiative voltage deficits originate from defective charge extraction at buried contacts or from trap-assisted recombination within bulk wide-bandgap (WBG) absorbers. Anchoring self-assembled monolayers (SAMs) directly to a reactive plasma deposition (RPD) tin oxide (SnOx) recombination layer to suppress contact barriers and halide accumulation delivers a certified power conversion efficiency (PCE) of 33.6% (champion 33.2% on 1 cm^2) and certified 31.0% on a 207.9 cm^2 minimodule [1]. In direct contradiction, competing reports conclude that carrier losses are dominated by photoinduced halide segregation and deep trap formation within the bulk WBG film, asserting that polymeric polycarbazole phosphonic acid (Poly-3Ph-4PACz) suppresses bulk recombination to yield a certified two-terminal (2T) tandem PCE of 32.95% (champion 26.13%, stabilised 27.12% in single junctions; 22.22% on a 57.6 cm^2 flexible module) [2]. Standard current-voltage scans cannot resolve this conflict because contact recombination velocity and bulk quasi-Fermi level splitting collapse produce identical open-circuit voltage (Voc) deficits. Decoupling contact resistance from bulk non-radiative recombination requires operando, spatially resolved voltage-dependent photoluminescence (PL) imaging via the local transport and recombination study (LTRS) method paired with bias-dependent transient absorption spectroscopy under continuous illumination to isolate interfacial from bulk carrier losses [138].

A second unresolved contradiction governs operational degradation at perovskite interfaces, where studies diverge on whether hole-selective contact decay proceeds via vertical molecular detachment or lateral dewetting, and whether bulky passivators relieve lattice strain or accelerate fatigue. One investigation attributes contact failure to out-of-plane molecular desorption that planar conjugated backbones suppress [163], whereas another concludes that degradation is driven by electric-field-induced lateral migration of self-assembled monolayer (SAM) molecules into isolated clusters [165]. Compounding this ambiguity, bulky organic passivators designed to raise phase transition barriers, such as 3-PMPCl reaching a certified 27.6% PCE in formamidinium lead triiodide (FAPbI₃) [159], induce local lattice expansion under thermal stress at heterointerfaces, driving interfacial fatigue even while unencapsulated devices retain 87% initial PCE after 4092 h of maximum power point tracking (MPPT) under International Summit on Organic Photovoltaic Stability (ISOS) protocol ISOS-L-1 with a time to 80% of initial performance (T80) of 3566 h [162]. Deciding whether contact failure originates from covalent bond cleavage, lateral dewetting, or thermal expansion mismatch requires operando bias-dependent sum-frequency generation spectroscopy coupled with in situ high-temperature grazing-incidence wide-angle X-ray scattering under continuous electrical load.

A third research gap concerns conflicting kinetic pathways for stabilizing the photoactive black alpha phase of FAPbI₃ against polymorphic transition into the photoinactive yellow delta phase. Contact-triggered cationic interaction asserts that phase stability occurs through non-destructive physical contact between separately crystallized two-dimensional (2D) and three-dimensional (3D) perovskites without interdiffusion, yielding a certified 25.61% PCE (champion 26.25%) [98]. In direct contrast, recrystallization using formamidinium tetrafluoroborate demonstrates that forming stable mixed-dimensional networks requires active formamidinium/methylammonium cation exchange, while 2D perovskitoid interlayers disrupt delta-phase nucleation to achieve an uncertified champion 25.61%

PCE [100]. Concurrently, low-dimensional passivators reach certified efficiencies of 26.2% (champion 26.49%) [19], but whether these capping layers introduce resistive transport bottlenecks remains unproven. Resolving whether phase stabilization proceeds by passive interfacial templating or dynamic chemical exchange requires in situ time-resolved grazing-incidence X-ray scattering paired with operando nanoscale secondary ion mass spectrometry during thermal annealing and light soaking.

A fourth mechanistic gap lies in opposing kinetic models of precursor-solvent coordination during scalable meniscus coating. One framework contends that prolonged solvent-solute interaction during blade coating stabilizes detrimental solvent-adduct intermediates, demanding rapid adduct suppression to achieve 26.5% PCE on 0.0665 cm^2 [186]. Conversely, competing investigations conclude that strong coordination by N-methyl-2-pyrrolidone or dimethyl sulfoxide is indispensable to retard premature nucleation and stabilize intermediate complexes, improving Voc from 1.08 to 1.09 V and fill factor (FF) from 75% to 80.5% [187], or achieving certified 26.22% PCE on 0.06 cm^2 through polypropylene glycol microcrystal regulation [194]. However, these coordination models remain demonstrated only on sub- 0.1 cm^2 apertures without verifying that intermediate equilibria persist across large areas under industrial drying and mechanical shear [195]. Closing this gap requires in situ optical reflectance and photoluminescence spectroscopy coupled with synchrotron X-ray diffraction during high-speed slot-die coating across modules exceeding 100 cm^2 .

Underpinning these physical gaps is a persistent empirical disconnect between certified peak efficiency, device area, and operational durability verification. Across the half year, the best certified PCE reached 34.85% in a 2T tandem, yet only 114 of 872 closely read papers reported any certified value. The best certified value moved from 29.2% in January to 33.6% in June, a change of 4.4 percentage points over the half year, while uncertified claims outpaced certified benchmarks, reaching 41.6% in April and 44.36% in May. Durability verification remains similarly sparse: only 39 papers reported a T80 lifetime, with the longest reaching 4100 h, and only 39 papers named an ISOS protocol. Although the largest reported aperture reached 900 cm^2 , champion efficiencies remain confined to sub- 0.1 cm^2 cells that mask sheet resistance, interconnect losses, and drying gradients. Crucially, this half year’s evidence could not support a rigorous comparison between buried contact passivators and bulk wide-bandgap stabilizers because reported devices diverged across uncalibrated active areas and unstandardized aging regimes. Instead of unencapsulated room-temperature scans on micro-cells, the field must measure encapsulated devices under continuous operational load using standardized ISOS-L-2 or ISOS-L-3 protocols at $65\text{ }^\circ\text{C}$, reporting certified steady-state power output on apertures of at least 1 cm^2 alongside spatially resolved luminescence maps of local contact resistance.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

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